Module-valued local triple derivations: the involutive bimodule case

Autonomous mathematical investigation, lane 5

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Abstract

The self-valued questions in arXiv:1208.3093 were completely answered in later work: every local triple derivation on a real or complex JB^*-triple is continuous, and every local triple derivation on a complex JB^*-triple or a C^*-algebra is a triple derivation. The later paper retains the harder question with values in an arbitrary Banach triple module. We prove a new module-valued theorem. If A is a unital C^*-algebra and X is any unital involutive Banach A-bimodule, then every bounded local triple derivation from A to X is a triple derivation for either of the two standard ternary module structures. This contains the previously known iterated-dual cases and applies to arbitrary involutive operator, dual, or reflexive bimodules. The proof combines orthogonal forms on commutative C^*-algebras, an ordered Arens-extension argument, support projections, generalized derivations, and Johnson's Jordan-derivation theorem. Arbitrary Jordan triple modules need not have the involution, bimodule products, or lifting-of-orthogonality property used by the proof, so the full module problem remains open.

1 The literature boundary

The original Problem 1 asks whether every local triple derivation on a JB^*-triple E, or more generally every local triple derivation $E \to X$ into a Jordan Banach triple E-module, is a triple derivation. It also asks the self-valued C^*-algebra question and whether local triple derivations are automatically continuous.

Burgos, Fernández-Polo, and Peralta subsequently proved that every bounded local triple derivation on a complex JB^*-triple is a triple derivation, that every local triple derivation on a real or complex JB^*-triple is continuous, and consequently that every local triple derivation on a C^*-algebra is a triple derivation [2]. Their Theorems 2.8 and 3.1 therefore settle all the self-valued and continuity questions from the source. However, their Problem 2.7 explicitly preserves the continuous module-valued question. Niazi and Miri later proved the positive answer for $A \to A^{(n)}$, for every iterated dual of a unital C^*-algebra [6]. They describe this as a partial answer.

The theorem below removes the iterated-dual restriction for the standard ternary structures induced by an involutive bimodule.

2 Two standard ternary module structures

Let A be a unital C^*-algebra. A *unital involutive Banach A-bimodule* is a unital Banach A-bimodule X with a bounded conjugate-linear involution $x \mapsto x^\#$ satisfying

$$(axb)^\# = b^* x^\# a^* \qquad (a, b \in A,\ x \in X).$$

An equivalent norm makes the involution isometric, but this is not needed.

There are two standard ternary A-module structures on X. In type II, put

$$[x,a,b]_{\mathrm{II}} = [b,a,x]_{\mathrm{II}} = \tfrac{1}{2}(xa^*b + ba^*x), \tag{1}$$

$$[a,x,b]_{\mathrm{II}} = \tfrac{1}{2}(ax^{\#}b + bx^{\#}a). \tag{2}$$

In type I, put

$$[x,a,b]_{\mathrm{I}} = [b,a,x]_{\mathrm{I}} = \tfrac{1}{2}(xab^* + b^*ax), \tag{3}$$

$$[a,x,b]_{\mathrm{I}} = \tfrac{1}{2}(a^*x^{\#}b^* + b^*x^{\#}a^*). \tag{4}$$

The bimodule identities show directly that these satisfy the ternary module axioms. A derivation into a type-II module is linear; a derivation into a type-I module is conjugate linear. For $A^{(n)}$, these are precisely the even and odd formulas, respectively, in [6, Proposition 2.4].

Theorem 1 (Involutive-bimodule local theorem)**.** *Let A be a unital C^*-algebra and X a unital involutive Banach A-bimodule. Equip X with either* (1)–(2) *or* (3)–(4)*. Then every bounded local triple derivation $T : A \to X$ is a triple derivation.*

No duality, reflexivity, normality, or representability assumption is made on X.

3 The commutative reduction

We write the JB^*-triple product on A as

$$[a,b,c] = \tfrac{1}{2}(ab^*c + cb^*a).$$

The following lemma is the main step. It is a bimodule-valued version of the commutative argument in [1, 6].

Lemma 2 (Commutative local lemma)**.** *Let C be a unital commutative C^*-subalgebra of A, and let $T : C \to X$ be a bounded local triple derivation with $T(1) = 0$. Then*

1. *in type* II*, T is an associative derivation $C \to X$;*
2. *in type* I*, $T \circ *$ is an associative derivation $C \to X$.*

Proof. First take $a,b,c \in C$ with $a^*b = b^*c = 0$. Choose a triple derivation $D_b : C \to X$ with $D_b(b) = T(b)$. In either standard module, the formulas show

$$[D_b(a), b, c] = 0 = [a, b, D_b(c)].$$

Also $[a,b,c] = 0$, so the derivation identity gives

$$[a, T(b), c] = 0. \tag{5}$$

Two applications of the orthogonal-form theorem for commutative C^*-algebras now give, for all $x,y,z,w \in C$,

$$[x, T(yz), w] = [x, T(y), z^*w] + [xy^*, T(z), w] - [xy^*, T(1), z^*w]. \tag{6}$$

For completeness, apply a functional $\varphi \in X^*$ to $[x, T(yz), w]$. Equation (5) says that the resulting sesquilinear form vanishes on orthogonal pairs. Goldstein's theorem represents it by a functional of a product. Applying the same theorem to the remaining pair of variables yields (6); Hahn–Banach removes φ. This proof uses only the module brackets, not any dual description of X.

We next extend (6) to C^{**} and X^{**}. The canonical Arens extensions of the left and right actions make X^{**} a unital involutive C^{**}-bimodule; the formulas above persist when each product is interpreted by successive duality. (Associativity follows by evaluating against X^* and taking weak-star approximants from C.) Take these extensions in the fixed order: the third outer variable, then the variable entering T^{**}, and finally the first outer variable. Equality of the bounded multilinear maps in (6) implies equality of these ordered extensions. Pairing successively with X^* shows explicitly that, for $y \in C$ and $x, z, w \in C^{**}$,

$$[x, T^{**}(yz), w] = [x, T(y), z^*w] + [xy^*, T^{**}(z), w] - [xy^*, T(1), z^*w]. \tag{7}$$

This ordered extension is why no separate weak-star continuity assumption on the original module actions is required.

Let $p \in C^{**}$ be the support projection of $b \in C$. In (7), take $y = b$, $z = p$, and $x = w = 1 - p$. Since $bp = b$, $p(1-p) = 0$, and $(1-p)b^* = 0$, all terms on the right vanish. For both types, $r = r^* = r^2$ implies $[r, \xi, r] = r\xi^{\#}r$. Hence

$$(1-p)T(b)^{\#}(1-p) = 0. \tag{8}$$

If $a^*b = b^*c = 0$, then $ap = pc = 0$, and (8) yields

$$aT(b)^{\#}c = 0. \tag{9}$$

The same argument for b^* gives $aT(b^*)^{\#}c = 0$.

Define a bounded linear map $G : C \to X$ by

$$G(q) = \begin{cases} T(q^*)^{\#}, & \text{type II}, \\ T(q)^{\#}, & \text{type I}. \end{cases}$$

In a commutative C^*-algebra, $ab = 0$ is equivalent to $a^*b = 0$. Thus (9) and its b^* version show

$$aG(b)c = 0 \qquad (ab = bc = 0).$$

The annihilator-preserving characterization of Li and Pan [5, Corollary 2.9] says that G is a generalized derivation. Since $G(1) = 0$, it is an associative derivation.

In type II,

$$T(yz) = G(z^*y^*)^{\#} = yT(z) + T(y)z,$$

so T is a derivation. In type I, the same computation gives

$$(T \circ *)(yz) = y(T \circ *)(z) + (T \circ *)(y)z.$$

This proves the lemma. $\square$

4 Normalization, symmetry, and the global proof

For either type and every $x \in X$, the module identity implies that

$$I_x(a) := [x, 1, a] - [1, x, a]$$

is an inner triple derivation. If T is local, a witness at 1 gives

$$T(1) = 2T(1) + T(1)^{\#}, \qquad \text{hence} \qquad T(1)^{\#} = -T(1).$$

Therefore

$$T_0 = T - I_{T(1)/2} \tag{10}$$

is again a bounded local triple derivation and $T_0(1) = 0$.

We need one more consequence of locality.

Lemma 3 (Symmetry after normalization)**.** *If $T : A \to X$ is a bounded local triple derivation with $T(1) = 0$, then*

$$T(a^*) = T(a)^{\#} \qquad (a \in A)$$

for either standard type.

Proof. Let $u \in A$ be unitary and restrict T to the commutative algebra $C^*(1, u)$. In type II, Lemma 2 and $0 = T(u^*u)$ give

$$T(u^*) = -u^*T(u)u^*. \tag{11}$$

Locality at u gives, using $[u, u, u] = u$,

$$T(u) = 2T(u) + uT(u)^{\#}u,$$

and the involution of this equality agrees with (11). Thus $T(u^*) = T(u)^{\#}$.

In type I, Lemma 2 applied to $T \circ *$ gives

$$T(u) = -u^*T(u^*)u^*.$$

Locality at u now gives

$$T(u) = -u^*T(u)^{\#}u^*,$$

so again $T(u^*) = T(u)^{\#}$. Every element of a unital C^*-algebra is a linear combination of unitaries. Taking account of the linearity in type II and conjugate linearity in type I extends the identity to every $a \in A$. $\square$

Proof of Theorem 1. By (10), it is enough to treat $T(1) = 0$.

In type II, set $D = T$; in type I, set $D = T \circ *$. In either case $D : A \to X$ is linear. For every self-adjoint $h \in A$, apply Lemma 2 to $C^*(1, h)$. Since $h = h^*$, it gives

$$D(h^2) = D(h)h + hD(h).$$

Polarization on the self-adjoint part and complex linearity show that D is a bounded Jordan derivation. Johnson's theorem [4, Theorem 6.3] implies that D is an associative derivation. Lemma 3 further gives

$$D(a^*) = D(a)^{\#} \qquad (a \in A). \tag{12}$$

For type II, $T = D$. Expanding $D(ab^*c + cb^*a)$ with the associative Leibniz rule and (12) gives exactly

$$T[a, b, c] = [T(a), b, c]_{\mathrm{II}} + [a, T(b), c]_{\mathrm{II}} + [a, b, T(c)]_{\mathrm{II}}.$$

For type I, $T(a) = D(a^*)$. Applying D to

$$[a, b, c]^* = \tfrac{1}{2}(c^*ba^* + a^*bc^*)$$

and using (12) gives exactly

$$T[a, b, c] = [T(a), b, c]_{\mathrm{I}} + [a, T(b), c]_{\mathrm{I}} + [a, b, T(c)]_{\mathrm{I}}.$$

Thus the normalized T is a triple derivation. Finally, $I_{T(1)/2}$ is a triple derivation, so the original T is one as well. $\square$

5 Scope and the remaining obstruction

Theorem 1 includes the iterated duals $A^{(n)}$, but its codomain can instead be any involutive Banach bimodule. Examples include involutive operator bimodules, normal dual bimodules, reflexive involutive bimodules, and direct sums or closed involutive submodules of these. The proof also explains which extra structure is doing the work.

It does not solve Problem 2.7 of [2]. A general Jordan Banach triple module need not lift orthogonality: $a \perp b$ in E need not imply $[a, b, x] = 0$ for $x \in X$ [7, Section 2]. Consequently the first orthogonality identity (5) is not available in general. Even in modules where that identity holds, there may be no involution or left/right products with which (8) can be converted into the one-sided zero-product relation used by Li–Pan. These are structural obstructions, not continuity gaps.

The accompanying attempt record gives eight focused upgrades, including a split-null-extension reduction and finite-dimensional counterexample tests. Those tests support rigidity but do not replace the absent general module structure. The correct classification is therefore a substantial new partial result together with a complete literature answer for the self-valued questions.

6 Verification and source evidence

The accompanying script checks both ternary structures on random complex matrix bimodules. It verifies the triple Leibniz identity for the two endgame forms $T = D$ and $T = D \circ *$, the inner normalization map I_x, and the symmetry identities for unitary matrices. Across the recorded trials, the largest Frobenius-norm residual is below 10^{-12}. These computations audit the algebraic formulas; the orthogonal-form and support-projection steps above are exact and do not depend on numerics.

Figures 1–3 show the exact source question, the retained module problem, and the later self-valued answer.

References

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satisfying:

$$\delta\{a,b,c\} = \{\delta(a),b,c\} + \{a,\delta(b),c\} + \{a,b,\delta(c)\}, \tag{1}$$

for every $a, b, c \in E$. In the setting of JB*-triples, J.T. Barton and Y. Friedman proved that every triple derivation on a JB*-triple is continuous (cf. [1], see also [11]). A *local triple derivation* on E is a linear map $T : E \to E$ such that for each a in E there exists a triple derivation δ_a on E satisfying $T(a) = \delta_a(a)$.

Quite recently, Jordan Banach triple modules over a JB*-triple and triple derivations from a JB*-triple E to a Jordan Banach triple E-module X were introduced by B. Russo and the fourth author of this note. In [23] these authors provide necessary and sufficient conditions under which a derivation from E into X is continuous. We refer to [23] and [12] for the basic definitions and results on JB*-triples, Jordan Banach triple modules and triple derivations not included in this note. Following [23], a conjugate linear mapping $\delta : E \to X$ is a *triple or ternary derivation* whenever it satisfies the above identity (1). In particular, the dual, E^*, of a JB*-triple E, is a Jordan Banach triple E-module and every triple derivation from E into E^* is continuous (see [23, Corollary 15]). Furthermore, every triple derivation from a C*-algebra B to a Banach triple B-module is automatically continuous [23, Theorem 20]. A bounded conjugate linear operator $T : E \to X$ is said to be a *local triple derivation* if for each $a \in E$, there exists a triple derivation $\delta_a : E \to X$ satisfying $T(a) = \delta_a(a)$. Clearly, every triple derivation is a local triple derivation, while the reciprocal implication is an open problem.

Problem 1. *Is every local triple derivation on a JB*-triple E (or more generally, every local triple derivation from E into a Jordan Banach triple E-module) a triple derivation?*

In a Conference held in Hong-Kong in April 2012, M. Mackey announced a result establishing that, for each von Neumann algebra (and more generally, for every JBW*-triple, i.e. a JB*-triple which is also a dual Banach space), W, every local triple derivation $T : W \to W$ is a triple derivation (see [21, Theorem 5.11]). Actually, the arguments provided by Mackey are also valid to prove that every local triple derivation on a compact JB*-triple is a triple derivation. The proofs and techniques applied by M. Mackey in this result depend heavily on the particular structure of a JBW*-triple

Figure 1: Problem 1 in arXiv:1208.3093, including the module-valued alternative.

LOCAL TRIPLE DERIVATIONS 11

triple derivation of the symmetrized Jordan triple product $3 < a, b, c >:= \{a, b, c\} + \{c, a, b\} + \{b, c, a\}$. $\square$

In the light of the above corollary, it seems natural to consider the following problem:

Problem 2.6. *Is every bounded local triple derivation on a real JB*-triple a triple derivation?*

In a recent paper (see [34]), B. Russo and the third author of this note initiated the study of triple module-valued triple derivations on (real and complex) JB*-triples. Let E be a complex (resp. real) Jordan triple. We recall that a *Jordan triple E-module* (also called *triple E-module*) is a vector space X equipped with three mappings

$$\{.,.,.\}_1 : X \times E \times E \to X, \quad \{.,.,.\}_2 : E \times X \times E \to X$$

$$\text{and } \{.,.,.\}_3 : E \times E \times X \to X$$

satisfying the following axioms:

($JTM1$) $\{x, a, b\}_1$ is linear in a and x and conjugate linear in b (resp., trilinear), $\{a, b, x\}_3$ is linear in b and x and conjugate linear in a (resp., trilinear) and $\{a, x, b\}_2$ is conjugate linear in a, b, x (resp., trilinear)

($JTM2$) $\{x, b, a\}_1 = \{a, b, x\}_3$, and $\{a, x, b\}_2 = \{b, x, a\}_2$ for every $a, b \in E$ and $x \in X$.

($JTM3$) Denoting by $\{.,.,.\}$ any of the products $\{.,.,.\}_1$, $\{.,.,.\}_2$ and $\{.,.,.\}_3$, the Jordan identity (1) holds whenever one of the elements is in X and the rest are in E.

When the products $\{.,.,.\}_1$, $\{.,.,.\}_2$ and $\{.,.,.\}_3$ are (jointly) continuous we shall say that X is a *Banach (Jordan) triple E-module*. Henceforth, the triple products $\{\cdot,\cdot,\cdot\}_j$, $j = 1, 2, 3$, which occur in the definition of Jordan triple module will be denoted simply by $\{\cdot,\cdot,\cdot\}$ whenever the meaning is clear from the context.

It is obvious that every real or complex Jordan triple E is a *real* triple E-module. The dual space, E^*, of a complex (resp., real) Jordan Banach triple E is a complex (resp., real) triple E-module with respect to the products:

$$\{a, b, \varphi\}(x) = \{\varphi, b, a\}(x) := \varphi\{b, a, x\}, \text{ and, } \{a, \varphi, b\}(x) := \overline{\varphi\{a, x, b\}},$$

for all $\varphi \in E^*$, $a, b, x \in E$.

Let E be a complex (respectively, real) JB*-triple and let X be a triple E-module. We recall that a conjugate linear (respectively, linear) mapping $\delta : E \to X$ is said to be a *triple or ternary derivation* if

$$\delta\{a, b, c\} = \{\delta(a), b, c\} + \{a, \delta(b), c\} + \{a, b, \delta(c)\}.$$

A conjugate linear (respectively, linear) mapping $T : E \to X$ will be called a *local triple derivation* if for each a in E there exists a triple derivation $\delta_a : E \to X$ such that $T(a) = \delta_a(a)$.

Problem 2.7. *Is every continuous local triple derivation from a real or complex JB*-triple E into a Banach triple E-module a triple derivation?*

Figure 2: The later paper arXiv:1303.4569 retains the arbitrary module-valued statement as Problem 2.7.

not assumed a priori to be continuous, on E. Pick a triple derivation δ_x satisfying $\delta_x(x) = T(x)$. In this case, we have

$$2\phi T(x) + \overline{\phi T(x)} = 2\phi \{T(x), x, x\} + \phi \{x, T(x), x\}$$
$$= 2\phi \{\delta_x(x), x, x\} + \phi \{x, \delta_x(x), x\} = \phi\delta_x \{x, x, x\}.$$

It follows from the above that

$$2\Re\mathrm{e}\phi T(x) = \phi\delta_x \{x, x, x\} - \phi T(x) = \phi\delta_x \{x, x, x\} - \phi\delta_x(x) = \phi\delta_x(\{x, x, x\} - x) = 0,$$

because δ_x is a triple derivation on E. This shows that T is dissipative.

Theorem 2.8. *Every local triple derivation on a (real or complex) JB*-triple is continuous. Consequently, every local triple derivation on a JB*-triple is a triple derivation.* □

3. Local triple derivations and generalised derivations on C*-algebras

We begin this section with the following corollary which solves Problem 2 in [14].

Theorem 3.1. *Every local triple derivation on a C*-algebra is a triple derivation.* □

In [14], it is established that every local triple derivation on a unital C*-algebra is a triple derivation. The strategy to prove this result relies on the connections between (local) triple derivations and generalised derivations on unital C*-algebras in the sense introduced by J. Li and Z. Pan in [30]. We recall that a linear mapping D from a unital C*-algebra B to a (unital) Banach B-bimodule X is called a *generalised derivation* whenever the identity

Figure 3: Theorems 2.8 and 3.1 of arXiv:1303.4569 settle automatic continuity and all self-valued complex JB^*-triple/C^*-algebra cases.